

Dibbayajyoti Roy

Voice AI & Backend Systems Engineer • Real-Time Speech Pipelines • Rust • TypeScript

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SUMMARY

Engineer who owns a production speech-to-speech AI interviewer end to end: rebuilt it from a failing high-latency cascade to a unified speech-to-speech pipeline (AWS Bedrock Nova Sonic, Pipecat, LiveKit/WebRTC) achieving 70-130ms response latency with Hindi-English code-switching, then ran it in production for real users on mobile and low-bandwidth networks. Six merged pull requests in production Rust codebases (Meta's pyrefly; ReductStore, a time-series database), including concurrency-sensitive observability work under strict maintainer review. High-agency by record: found the problems, shipped the fixes, nobody assigned them. Eligible for EU Blue Card; ready to relocate to Berlin.

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer, Yupcha Softwares Pvt. Ltd. • Jul 2025 to Present • Agartala, India

- Rebuilt a production voice AI interviewer from an inherited cascaded STT/LLM/TTS pipeline into unified speech-to-speech on AWS Bedrock (Nova Sonic) with Pipecat over LiveKit (WebRTC); cut response latency to 70-130ms and eliminated session freezes on mobile and low-bandwidth connections. Owned it in production.
- Evaluated and integrated speech models for the Indian market, including Hindi-English code-switching behaviour, prompt and pipeline tuning, and interruption handling.
- Diagnosed a multi-track audio/video sync defect in interview recordings through three bug layers (LiveKit egress, FFmpeg filter graphs, double delay compensation via adelay and -itsoffset); shipped per-track compensation.
- Cut a hot Redis polling path ~90% by rewriting SCAN+GET as MGET batching; built a network simulation and diagnostics spec to reproduce poor-network call conditions deterministically.
- Shipped supporting product surfaces end to end: Next.js frontends (page load 3.4s to 1.9s via N+1 removal, composite indexes, bundle splitting), 12+ Node.js REST endpoints over PostgreSQL, Docker deployment on Linux/Proxmox with nginx and systemd.

OPEN SOURCE: SYSTEMS WORK UNDER MAINTAINER REVIEW (RUST)

- **ReductStore** (time-series database for robotics/IoT): five merged PRs in the observability layer, including a maintainer-requested cross-cutting refactor unifying all system event handling (#1496), reentrancy-safe log capture with task-local guards against infinite logging loops (#1467), and replication diagnostics shipped in v1.20 with public credit from the co-founder (#1417). Concurrency-safe design throughout: async locks, non-blocking telemetry on the write path.
- **pyrefly** (Meta's Python type checker): fixed a false-positive untyped-import diagnostic per PEP 561, correctly handling submodules (#3840).

TECHNICAL SKILLS

- **Voice & Real-Time:** AWS Bedrock (Nova Sonic), Pipecat, LiveKit, WebRTC, FFmpeg, Server-Sent Events; latency profiling and interruption handling
- **Languages:** TypeScript, Rust, JavaScript, SQL, Python
- **Systems & Concurrency:** async Rust (tokio): channels, async locks, task-local state, reentrancy-safe design; observability pipelines
- **Backend & Data:** Node.js, Hono, Express, REST, Zod, OAuth, JWT; PostgreSQL (RLS, partitioning, indexing), Redis, DynamoDB
- **Infrastructure:** Linux, Docker, Proxmox VE, nginx, systemd, AWS (Bedrock, DynamoDB, CloudWatch), Cloudflare Workers

SELECTED PROJECTS

- **whatbroke** (TypeScript CLI): crash-context packager for AI debugging; MCP server with six tools, git-anchored suspect ranking, parsers for five test runners; on npm.
- **AHTML** (TypeScript): agent-readable HTML endpoint standard emitting WebMCP, OpenAPI 3.1, JSON-LD, and llms.txt from one source; npm suite with 8.7k+ downloads.

HONORS & EDUCATION

Winner, NITA Arjuna 2.0 National Hackathon (2025), 200+ teams • Winner, Technovate Project Exhibition (2025) • 1st Runner-Up, NITA-ISRO Space Hackathon (2024)

B.Tech, Computer Science, ICFAI University Tripura (completed 2026; conferral October 2026)

English (professional), Bengali (native), Hindi (fluent) • Eligible for EU Blue Card / Germany Opportunity Card • Ready to relocate to Berlin